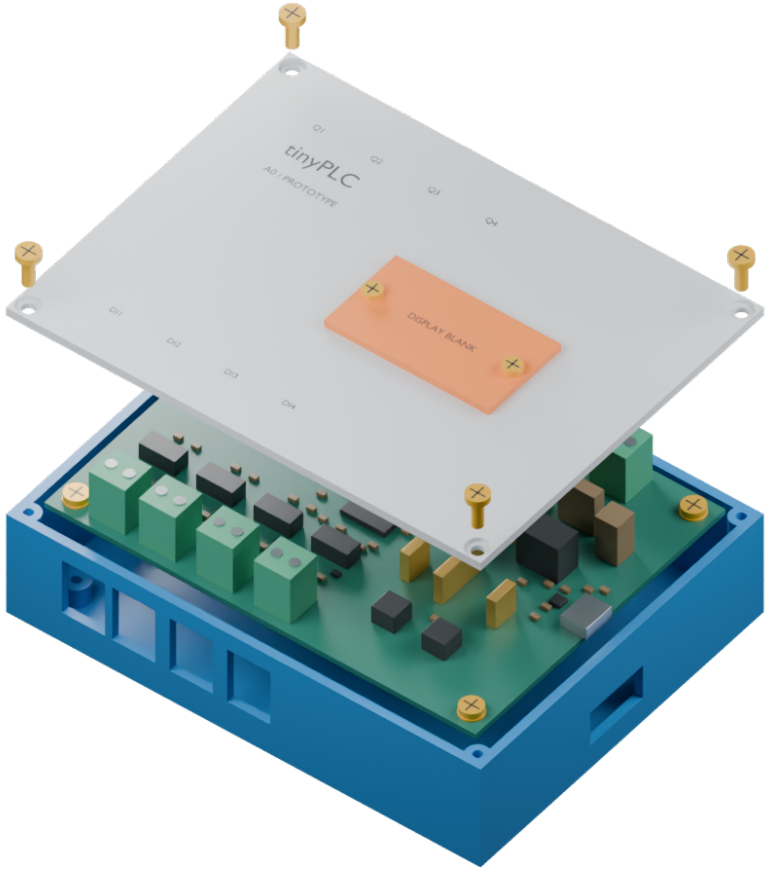
Build it. See every step.

YOUR FIRST
tinyPLC CASE

A picture-led guide to printing, checking and assembling the A0 enclosure.



PRINT + FIT NOW

Start with the coupon, printed case and card board template.

PCB INSTALLATION LATER

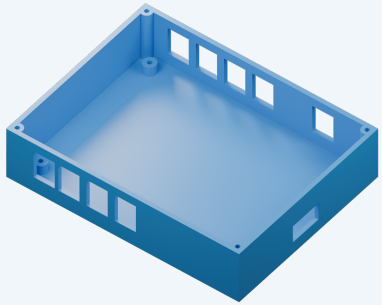
The A0 PCB is unrouted. Board steps show the intended future assembly.

Blue base • pale lid • orange blank • green PCB • yellow hardware

Colors identify parts. Hardware and electronic packages are simplified; screws are candidate sizes. Keep the detailed electrical procedure beside this booklet.

Lay out the large parts

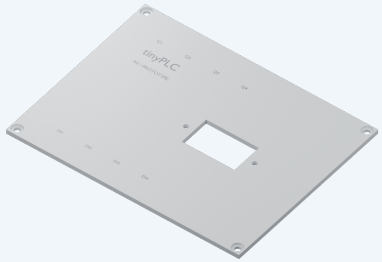
A BASE



1 × base-print.stl

Blue shell; integral PCB posts.

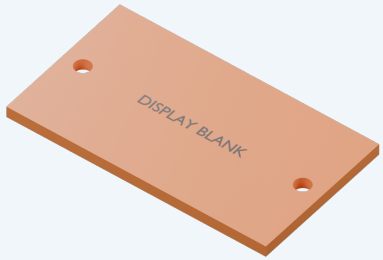
B LID



1 × lid-print.stl

Pale lid; locating lip underneath.

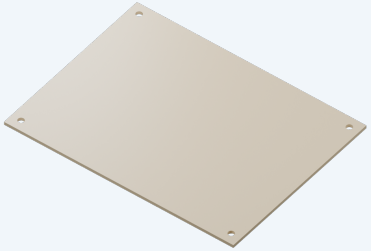
C DISPLAY BLANK



1 × display-blank-print.stl

Orange cover; no display installed.

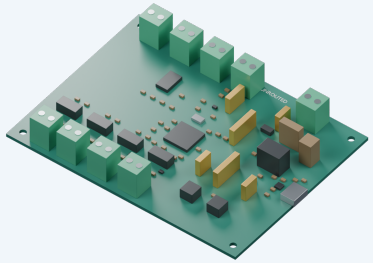
D CARD TEMPLATE



1 × drawing 03 at 100%

Use this for the first case fit.

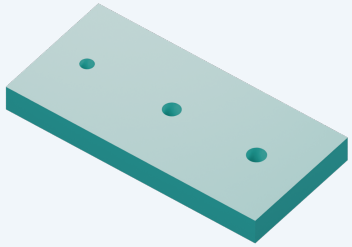
E FUTURE PCB



1 × reviewed, tested board

Shown for planning; not ready to order.

F FIT COUPON



1 × fit-coupon-print.stl

Check printer and pilot holes first.

Sort the screws before you start



4 ×

M3 × 6

PCB mounting

With 4 M3 washers, about 0.5 mm thick.



4 ×

M3 × 8

Lid mounting

Head diameter at most 6 mm.



2 ×

M2.5 × 8

Display blank

Pass through blank and lid.



4 ×

M3

Under PCB screw heads

Seat on the board, not below it.



2 ×

M2.5

Under the lid

Between lid underside and blank nuts.



2 ×

M2.5

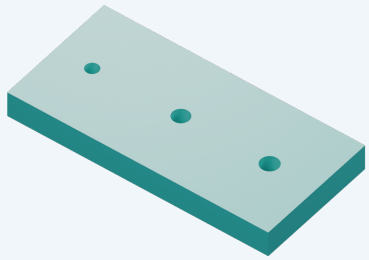
Display blank retention

Hold while tightening the two screws.

Tools: ruler, deburring tool, M3 × 0.5 hand tap, suitable hand screwdriver and a small nut driver. Confirm actual screw drawings and fit on the coupon; no tightening torque is qualified.

01 Prove the fit on the small coupon

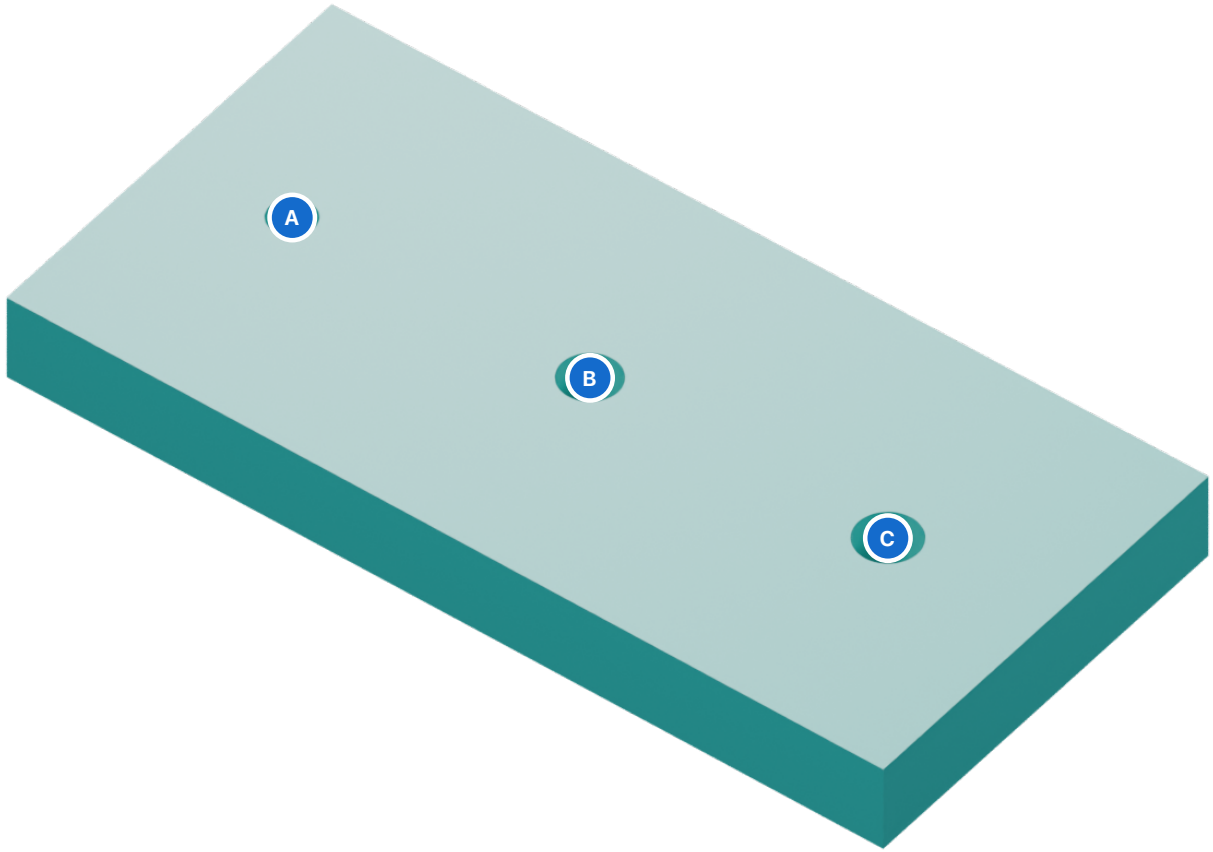
USE FIRST



1 × fit coupon

Print before the full case. Check the 50 × 24 mm outline with a ruler.

A: 2.5 mm pilot — try M3 tap. B: 3.2 mm clearance. C: 3.4 mm clearance.

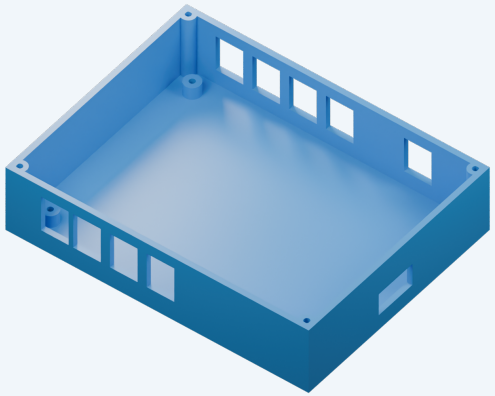


CHECK

No splitting, binding or stripping.

02 Print with the right face on the bed

BASE: FLOOR DOWN

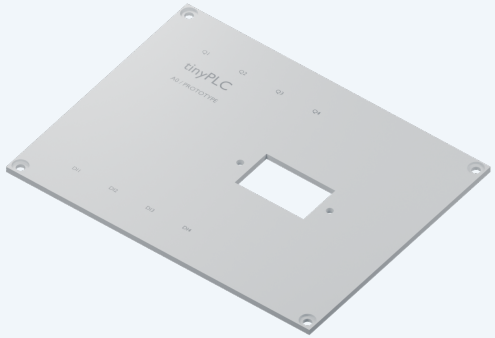


base-print.stl

Open cavity faces up.



LID: OUTSIDE FACE DOWN

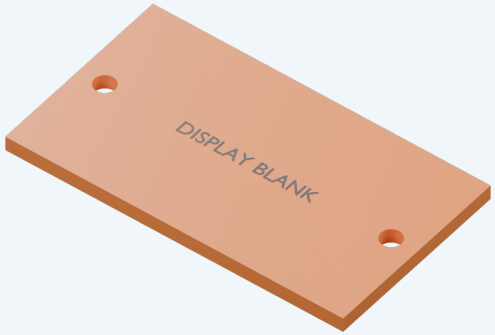


lid-print.stl

Locating lip faces up.



BLANK: FLAT



display-blank-print.stl

Both mounting holes remain clear.

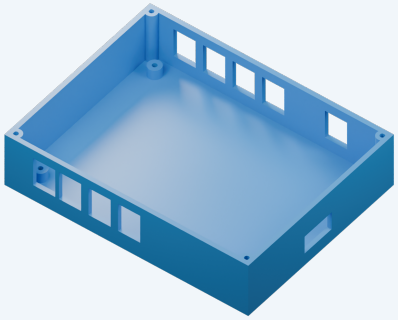


Use the supplied print STLs in millimeters: their print orientations are already applied. The pictures above identify parts; the small sections show bed orientation. Use the coupon settings that passed the previous step.

Starting proposal: PETG, 0.4 mm nozzle, 0.2 mm layers. Review bridges/supports in the slicer and record your actual profile.

03 Clean and prepare the blue base

USE THESE PARTS



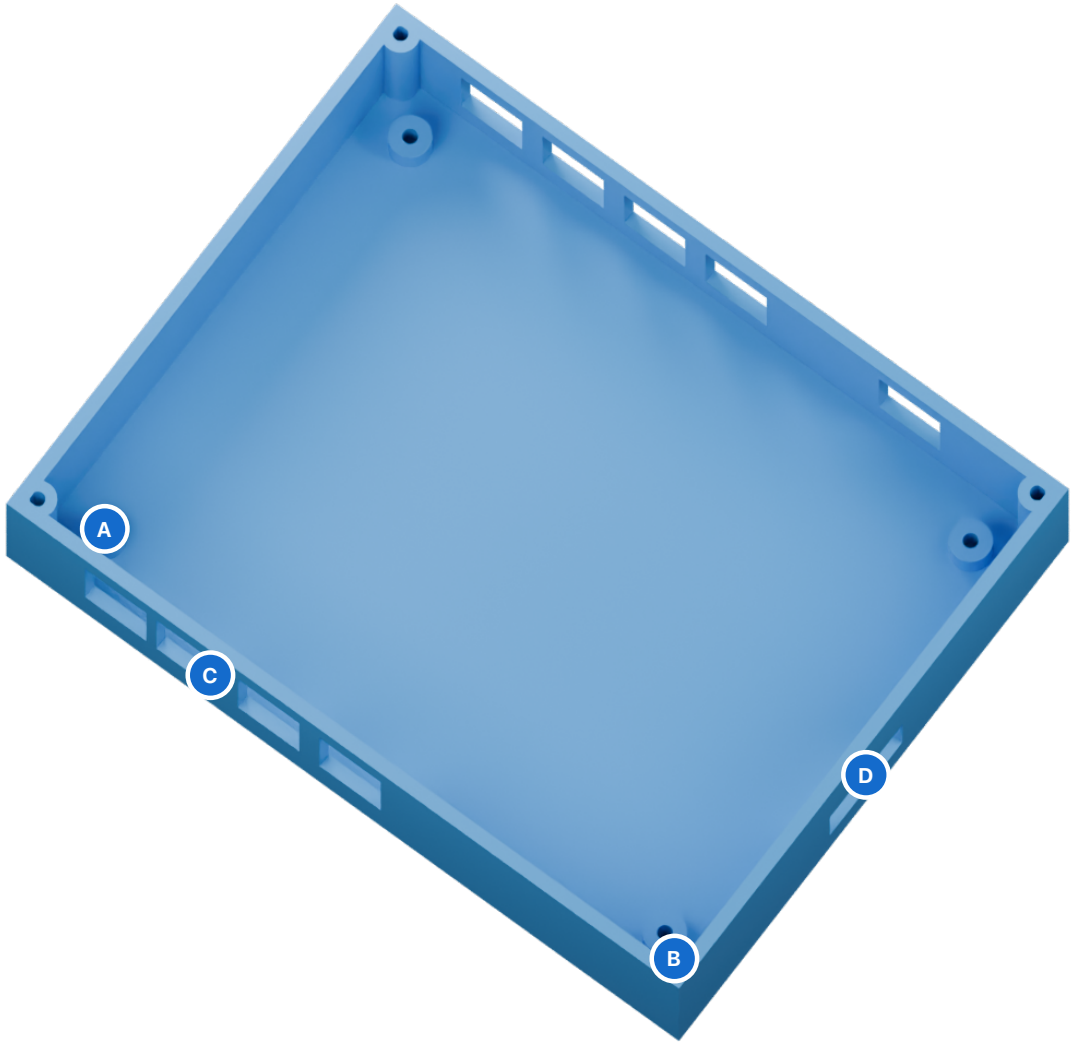
1 × base

Remove supports and strings. Try the empty lid; it must seat without screws pulling it flat.

A: four PCB posts. B: four lid towers. Tap their pilots only after the coupon passes.

CHECK

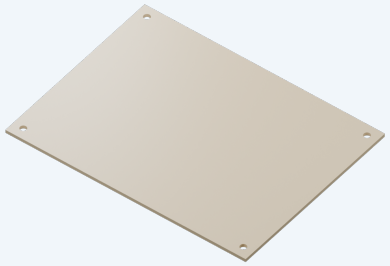
Pilots clear. Floor and lid flat.



C: DI wire openings D: USB service opening

04 Try the card board template

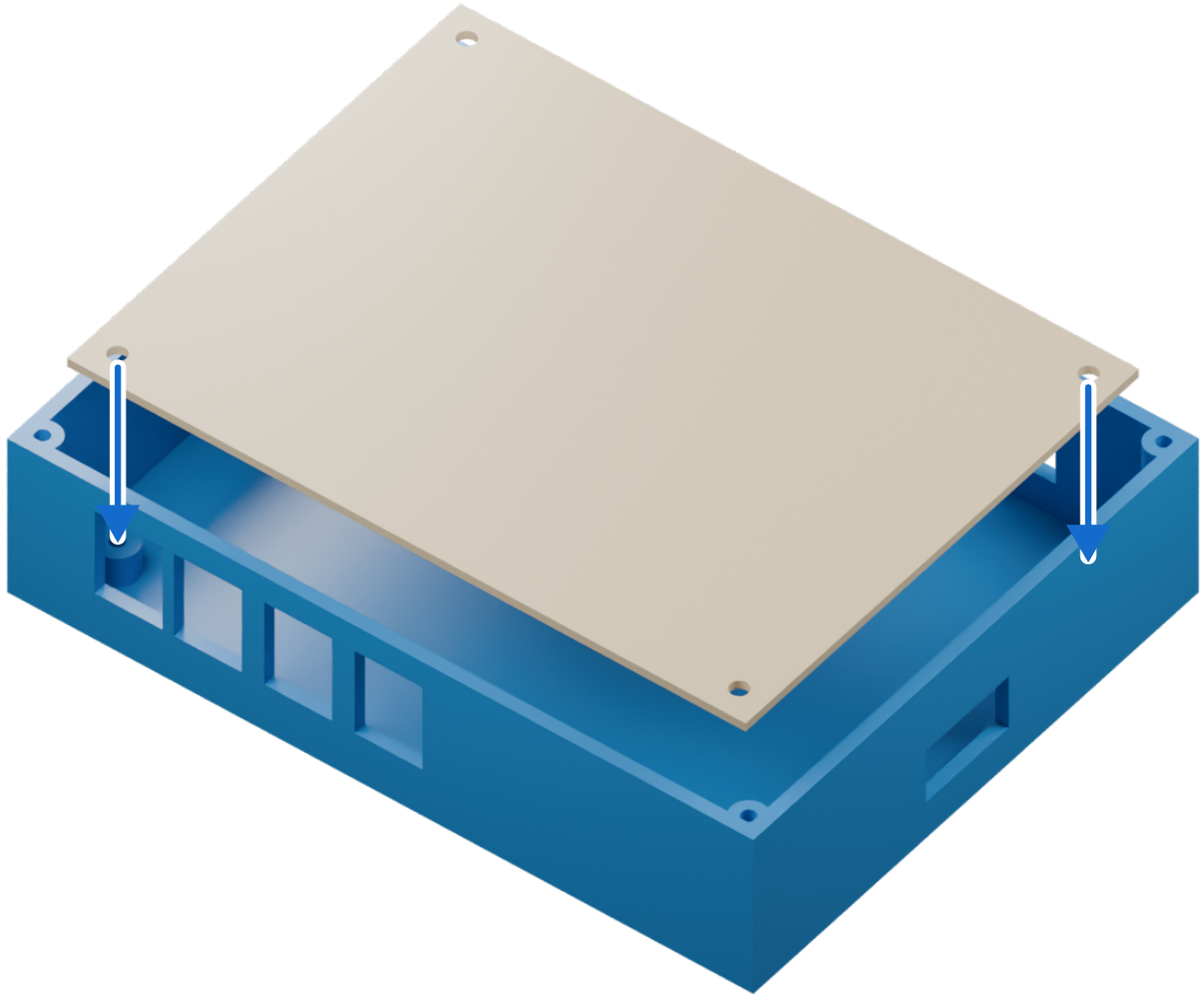
ADD THESE PARTS



1 × card template

Print drawing 03 at 100%. Check its 100 mm scale bar, cut the outline and punch four holes.

Lower onto the posts. DI edge faces the four front openings. Check USB and terminal positions.

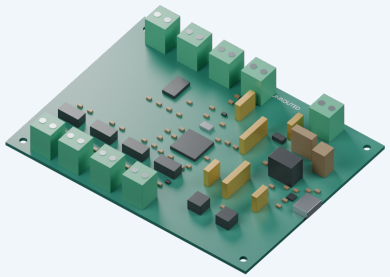


CHECK

All four holes align freely.

05 Lower the tested board into the base

ADD THESE PARTS



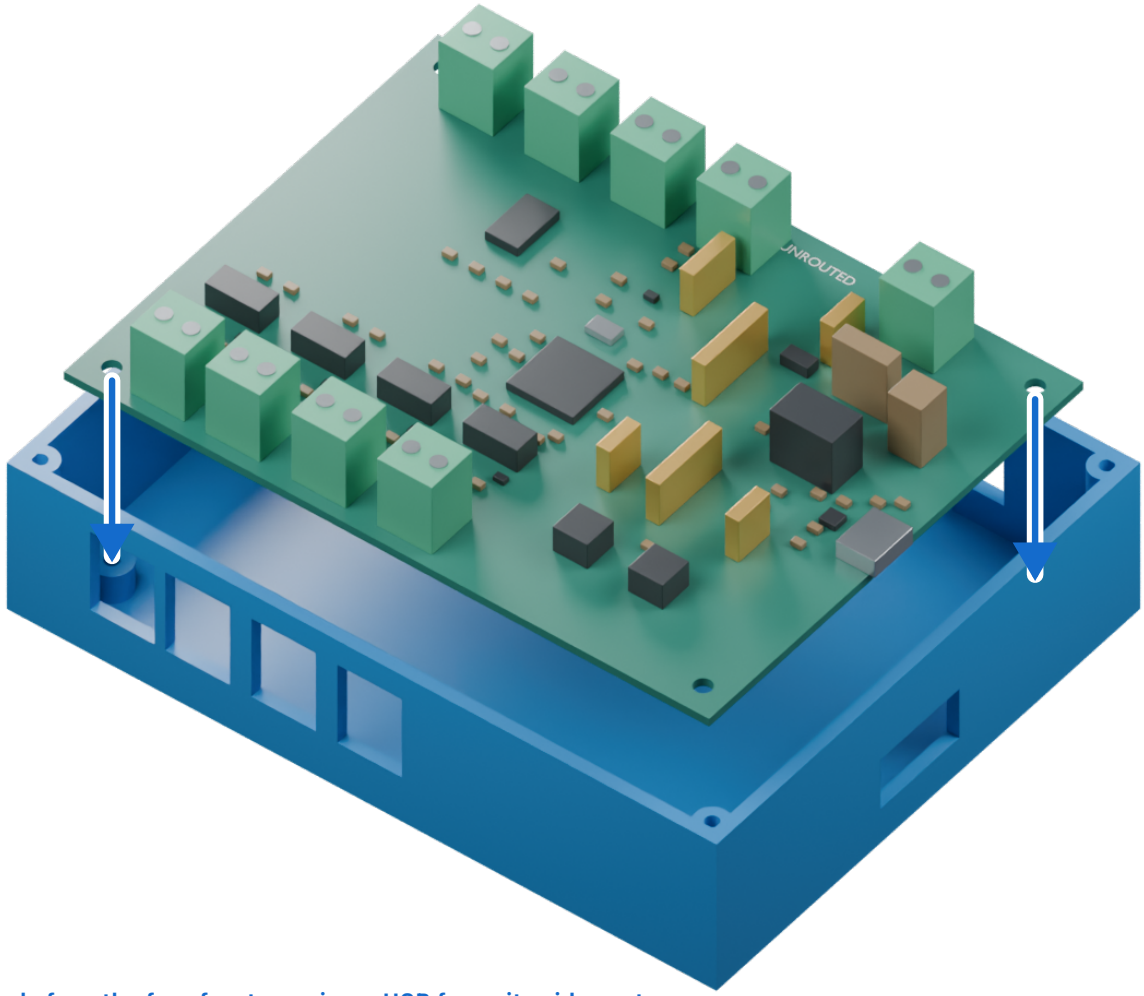
1 × tested PCB

First finish routing, fabrication review and the electrical bring-up in the technical reference.

Power off. Remove cables. Inspect underside leads, then lower the board onto all four posts.

CHECK

Board flat; no lead touches case.



DI terminals face the four front openings; USB faces its side port.

06 Secure the board with four screws

ADD THESE PARTS

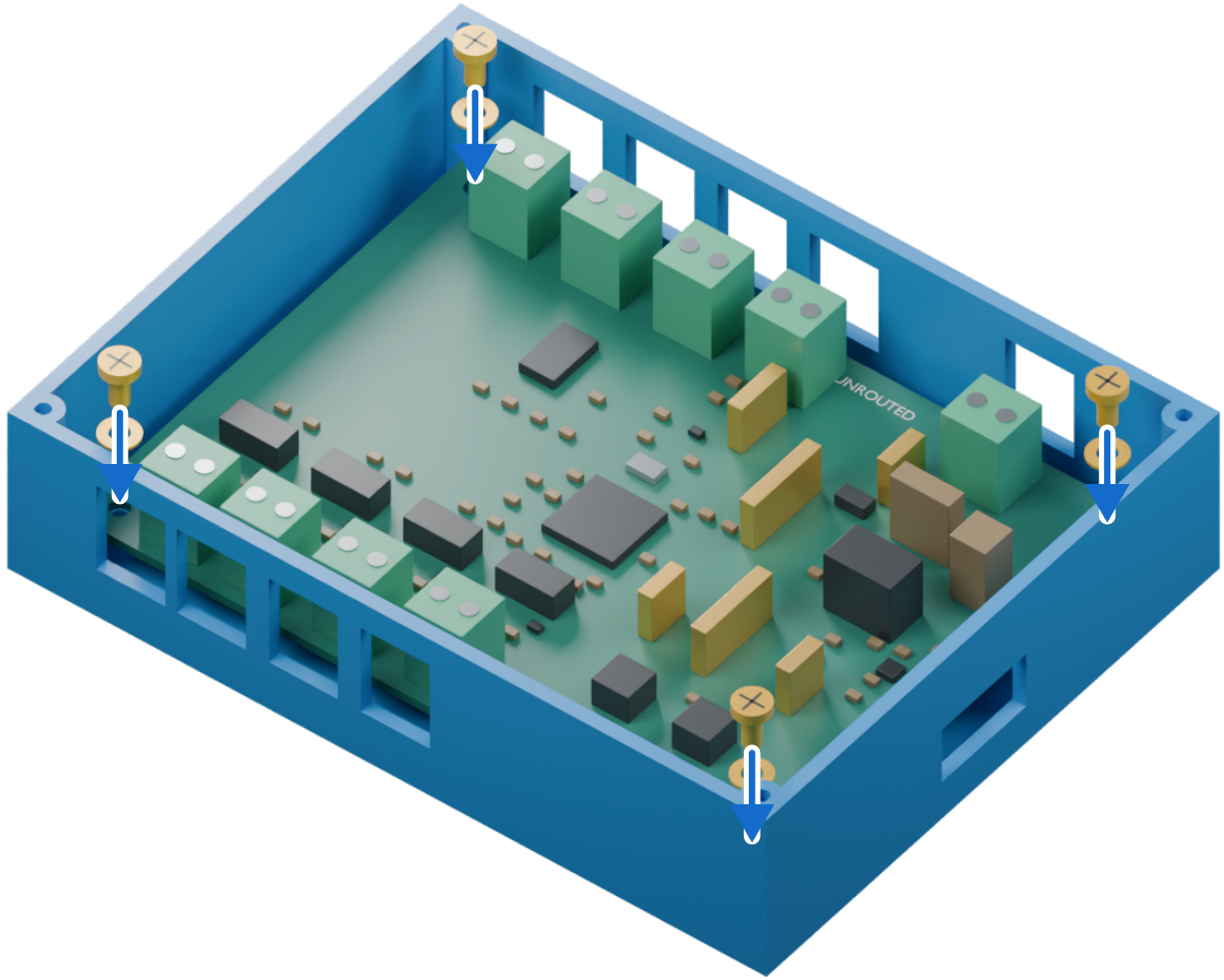


4 ×
M3 × 6



4 ×
M3 washer

Washer goes under each screw head, on top of the PCB. Tighten gently by hand in opposite corners.



CHECK

No bowing or screw bottoming.

07 Attach the orange blank to the lid

ADD THESE PARTS

1 × lid + 1 × blank



2 ×
M2.5 × 8

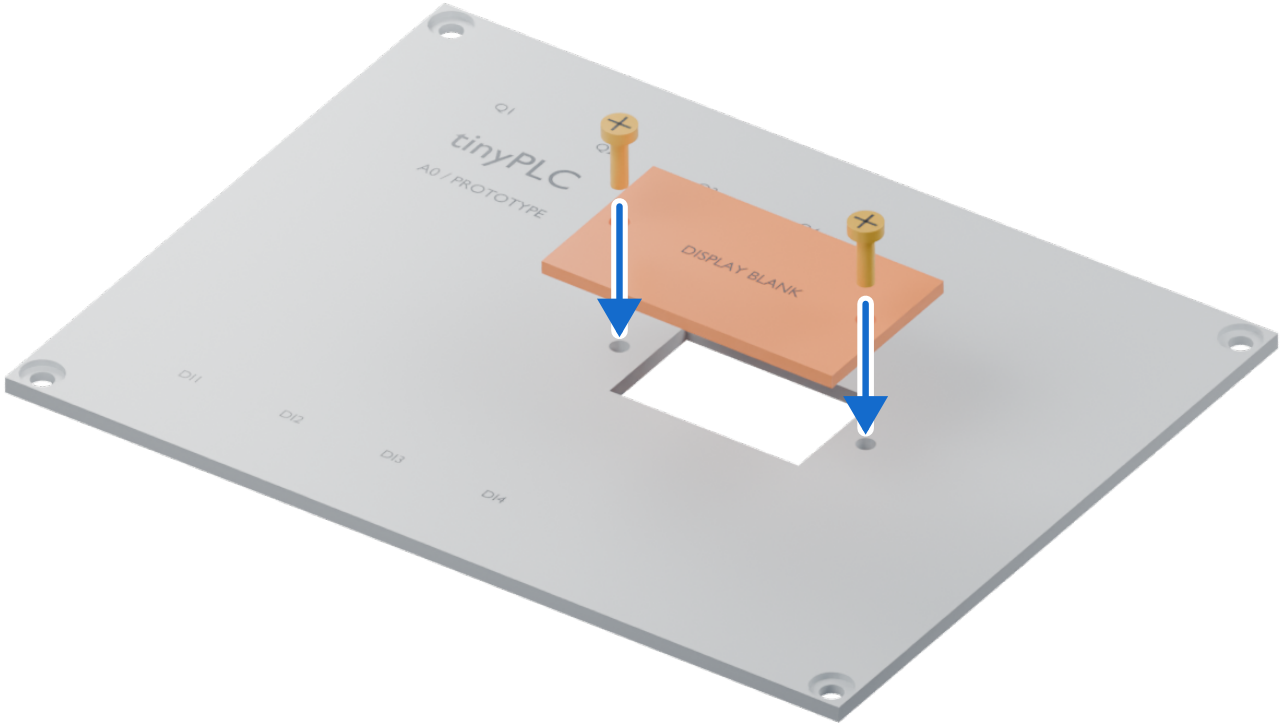


2 ×
M2.5 washer



2 ×
M2.5 nut

Screw → blank → lid → washer → nut.



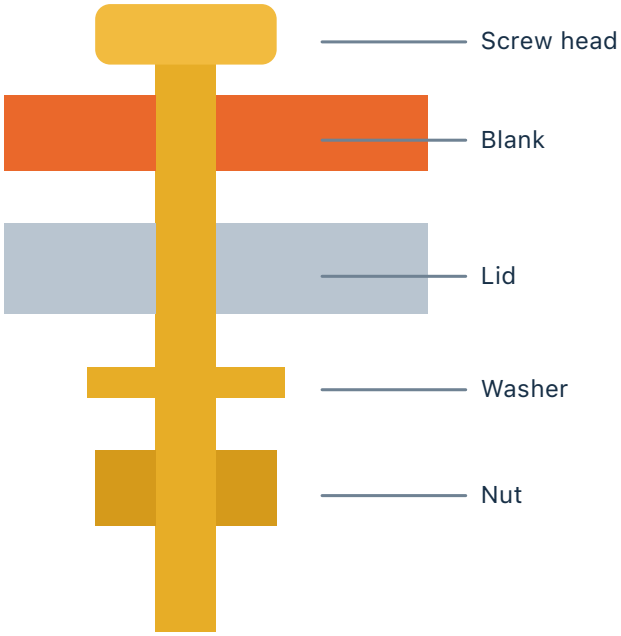
CHECK

Blank covers the entire opening.

Washers and nuts go underneath: see the detail on the next page.

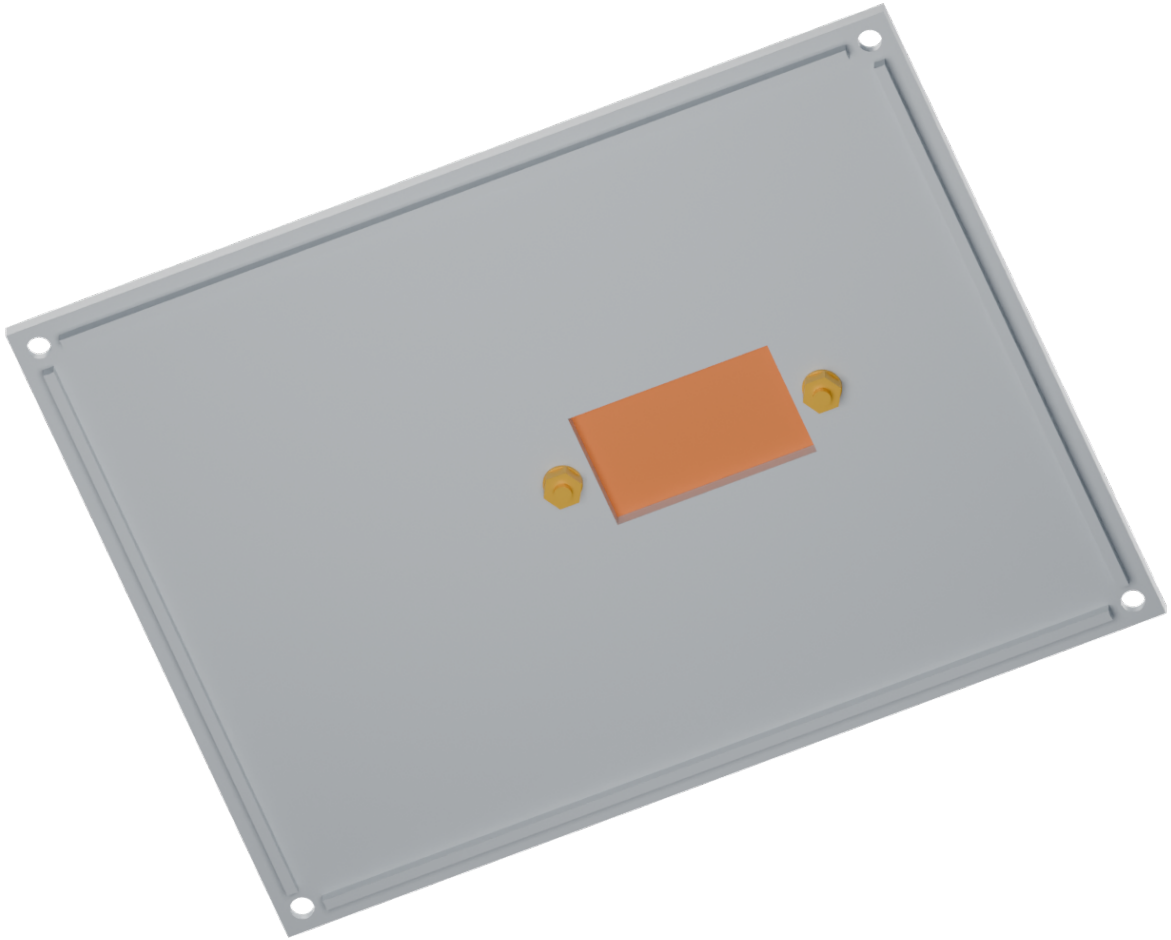
The blank fastener stack, from both sides

SIDE SECTION / ONE FASTENER



M2.5 × 8: length below the head

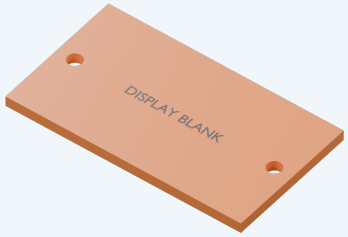
Section is illustrative; gaps are exaggerated to show the order. Washers belong below the lid.



UNDERSIDE VIEW / two nuts + two washers

08 Check the completed lid

COMPLETED SUBASSEMBLY



Hold each nut while tightening its screw. The blank should sit flat with no movement.

Check underneath: both washers and nuts are present. Keep them clear of future wiring.

CHECK

Two screws, two washers, two nuts.



Lid lettering is a separate applied label; it is not embossed in the STL.

09 Inspect the open assembly and wiring

CHECK BEFORE CLOSING

A All 4 board screws seated.

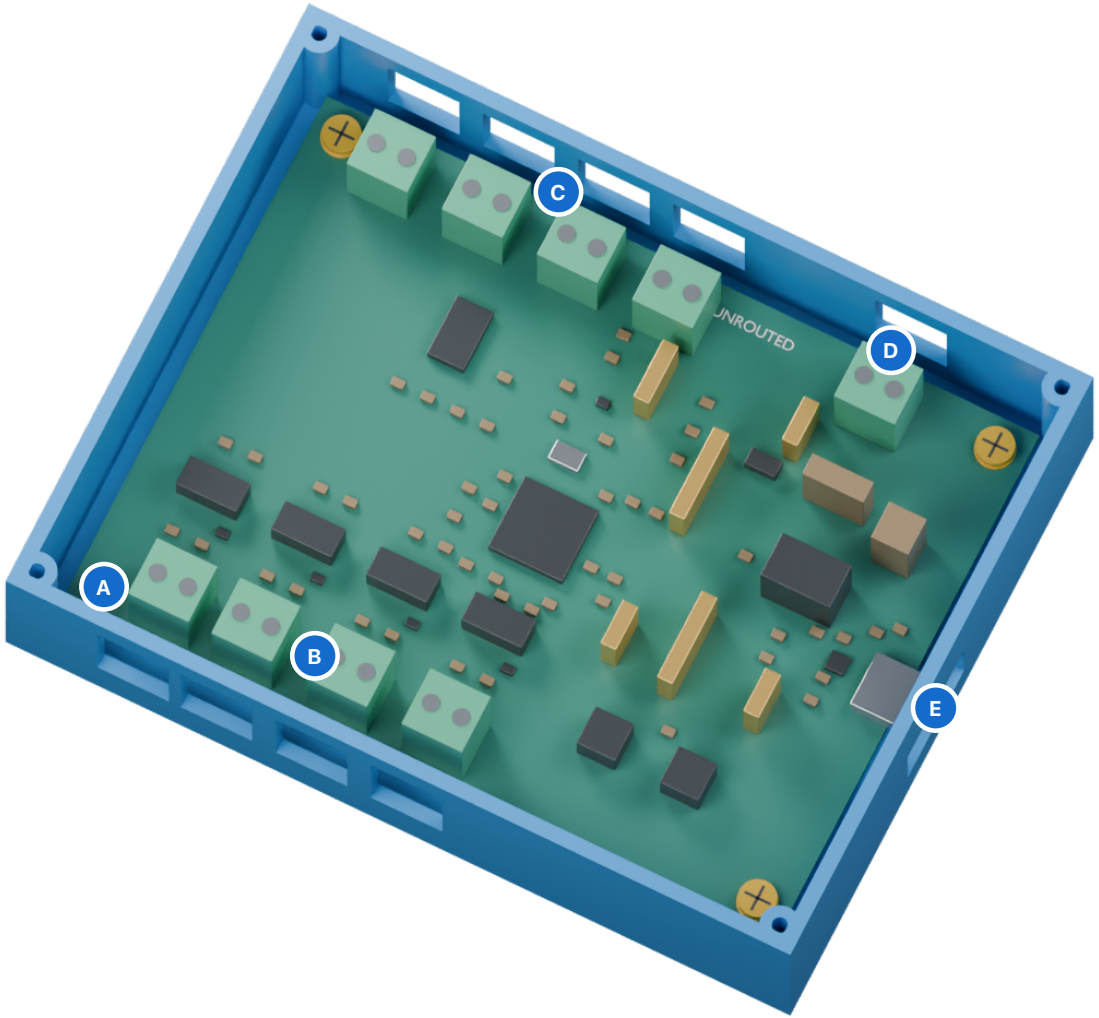
B DI1–DI4 at the front.

C Q1–Q4 at the rear.

D 24 V supply connector.

E USB service opening.

Use schematic pin numbers for polarity, never this picture alone.



CHECK

No loose hardware or clipped leads.

10 Seat the lid without trapping wires

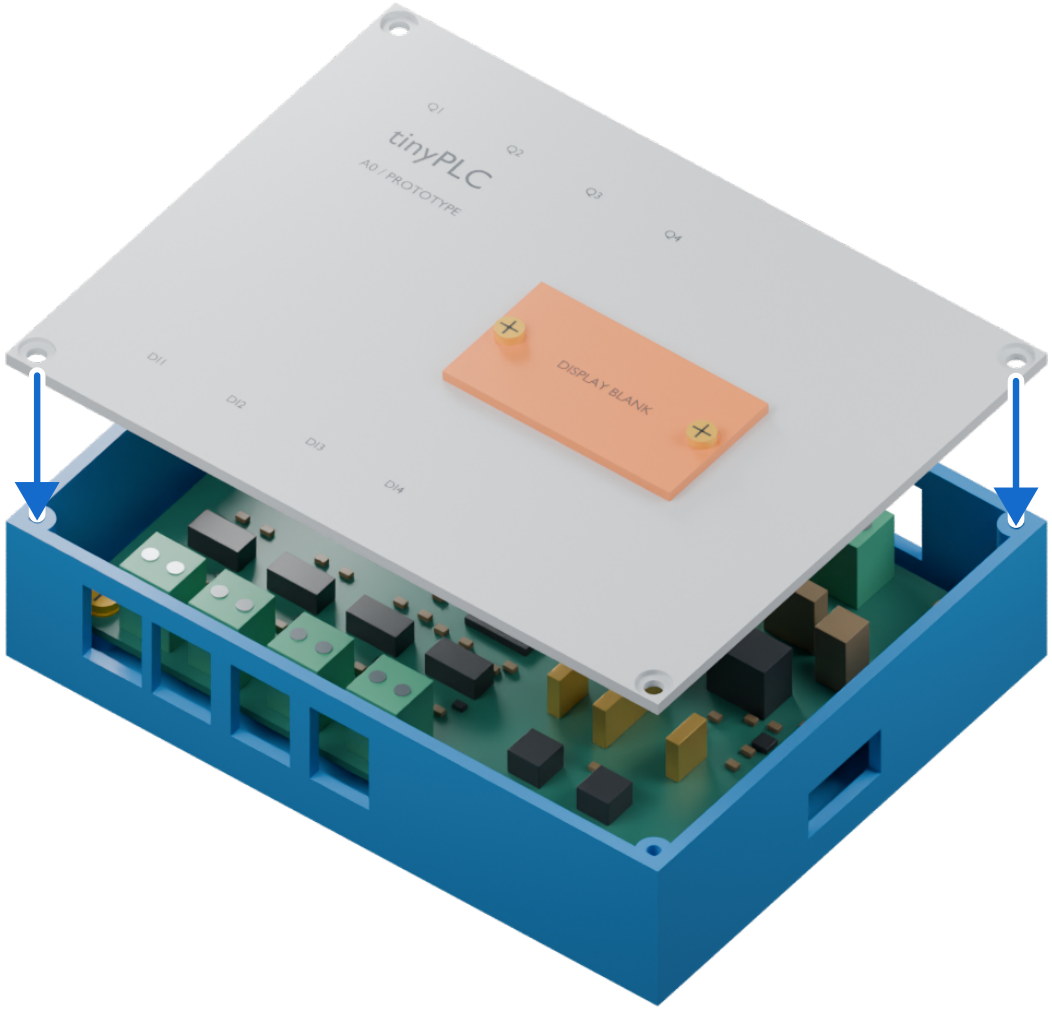
ADD THESE PARTS



1 × completed lid

If wired: label each wire, use the terminal maker's strip length and provide external strain relief.

Keep wires away from the lip and blank nuts. Lower the lid until the lip sits inside the case.



CHECK

Lid sits flush before screws go in.

11 Fit the four lid screws

ADD THESE PARTS

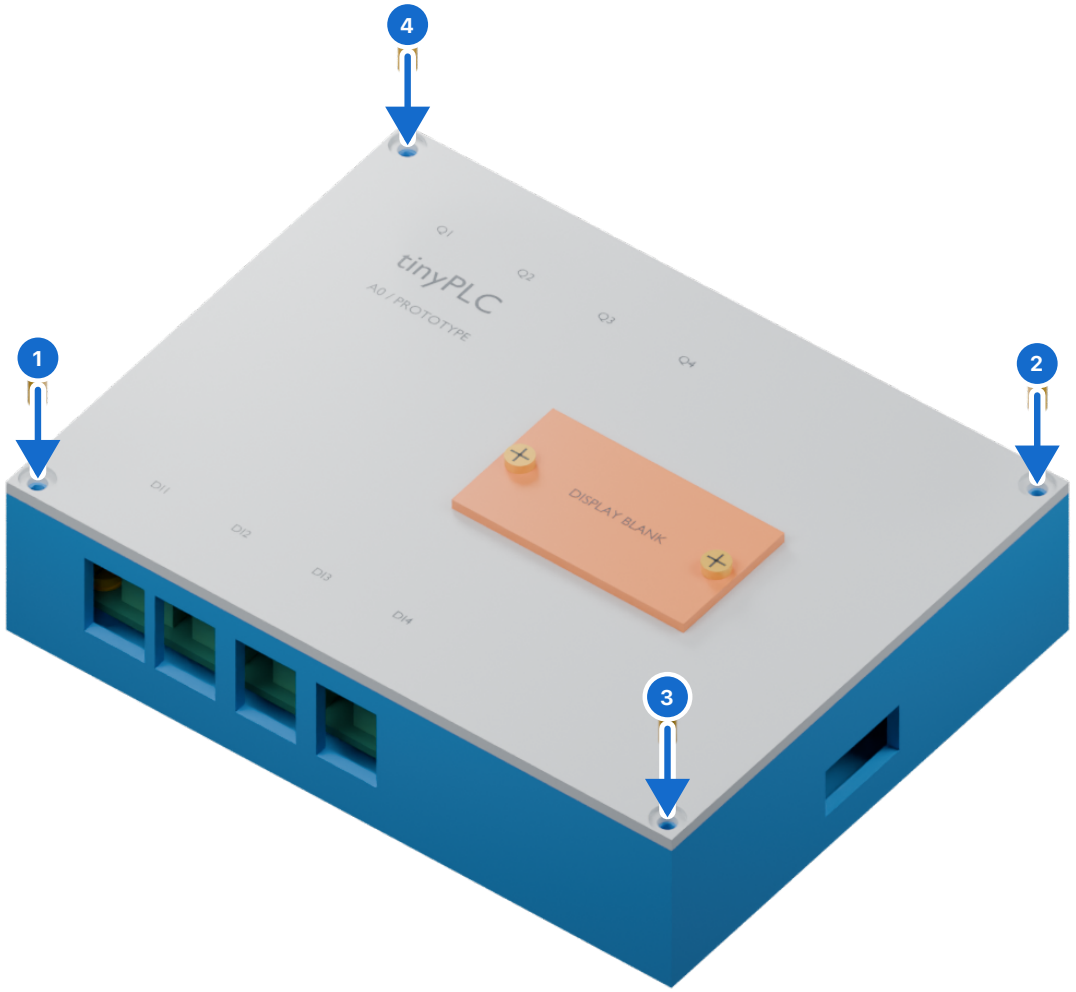


4 ×

M3 × 8

Start all four screws by hand. Seat them lightly in the illustrated order: 1 → 2 → 3 → 4.

Heads must fit the counterbores. Stop if a screw bottoms out or the plastic starts to distort.



CHECK

Even seam; no forced lid bending.

Record the fit. Plan the next revision.

FINAL CHECK

- [ ] Case seam is even.
- [ ] Blank is secure.
- [ ] Ports stay accessible.
- [ ] No trapped wiring.
- [ ] Labels match the schematic.

Record print settings, fit issues and measurements. Revise CAD and regenerate matching files.



Next: electrical bring-up and measured testing

Follow technical-assembly-reference.pdf for staged soldering, power checks, SWD, I/O and firmware. A closed case does not establish a working controller.